

MANDAR HIRPHODE

Data Analyst | BI & Analytics | Energy-Tech & ESG Intelligence

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PROFESSIONAL SUMMARY

Data Analyst with hands-on production experience in energy-tech and Building Management Systems (BMS). Specialises in IoT time-series analytics, real-time monitoring, ESG/IPMVP energy baseline modelling, and end-to-end BI pipelines across Power BI, Grafana, and SQL. Delivered measurable outcomes for 5+ enterprise clients across energy, real estate, and smart-building domains. IIT Guwahati certified in Full Stack Data Science & AI. Currently expanding into Snowflake and PyTorch.

CORE SKILLS

BI & Visualisation: Power BI (DAX, Power Query), Grafana (Dashboards, Alerting), Tableau, Looker Studio, Advanced Excel
Databases & SQL: PostgreSQL, InfluxDB (Flux, InfluxQL), MySQL, MS SQL Server, Snowflake (learning)
Programming & ETL: Python (Pandas, NumPy, Matplotlib, Scikit-learn), ETL Scripting, Microsoft Fabric, SSIS, Pipeline Automation
Analytics & ML: IPMVP Regression, Energy Forecasting, Time-Series Analysis, Clustering, EDA, Predictive Modelling
Domain & Tools: BMS Systems, ESG/ASHRAE Standards, Azure, Dimensional Modelling, Real-time IoT Ingestion | Learning: Snowflake, PyTorch

PROFESSIONAL EXPERIENCE

Data Analyst — Integrated Active Monitoring, Pune Jan 2025 – Present

- Built real-time IoT monitoring dashboards in Grafana for BMS KPIs — energy, temperature, AQI (CO2, PM2.5), lift utilities, and MCB status across 5+ enterprise clients; configured threshold-based alerting to notify clients via email/phone.
- Developed IPMVP-compliant regression models in Python for energy baseline calculation using occupancy, CDD/HDD, and working-hour parameters — improving energy optimisation accuracy by 15%.
- Designed Power BI forecasting dashboards for energy consumption and peak demand prediction, reducing client penalty costs by 10%.
- Engineered PostgreSQL-Python-Grafana and PostgreSQL-Python-Power BI pipelines integrating InfluxDB time-series data, automating weekly/monthly reporting and saving 10+ manual hours per week.
- Built access control, footfall analytics, and login behaviour dashboards; contributed to ESG/ASHRAE product development by defining KPIs for the BMS web portal with the software team.
- Collaborated cross-functionally with embedded software, tech support, and project teams; delivered client-facing BI storytelling and custom reports to multiple enterprise stakeholders.

INTERNSHIP EXPERIENCE

Data Science Intern — RajYug IT Solutions Sept – Dec 2024

- Built Power BI and Tableau dashboards for Travel, Oil & Gas, and Real Estate sectors with advanced DAX and Power Query; developed ETL pipelines using Microsoft Fabric and SSIS.
- Built ML regression and classification models for demand forecasting and predictive maintenance, improving forecast reliability by 12%.

Data Analyst Intern — YBI Foundation May – July 2024

- Performed EDA, data cleaning, and clustering using Python; built churn prediction models and customer behaviour dashboards in Power BI.

KEY PROJECTS

- Croma Retail Data Warehouse** — End-to-end Snowflake + Azure warehouse with Snowpipe ingestion, dimensional modelling, and KPI analytics for sales, inventory & CLV.
- End-to-End Analytics Pipeline** — Python to SQL to Power BI pipeline delivering real-time business insights, boosting decision-making efficiency by 20%.
- Movie Recommender System** — Content-based recommender using TF-IDF vectorisation and cosine similarity with 85% accuracy.
- EDA Projects** — Google Play Store & IPL datasets — trend analysis, rating patterns, and match performance insights using Python.

EDUCATION & CERTIFICATIONS

Full Stack Data Science & AI — IIT Guwahati (2023–2024) | B.Tech — Shivaji University (2020–2023)

■ Employee of the Month — Integrated Active Monitoring (Nov 2025)